

Power sector transition dynamics

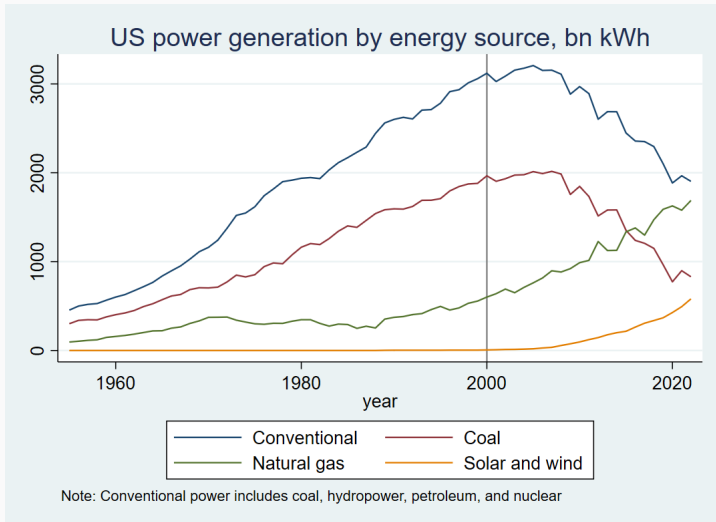
Maija Kaartinen and Alireza Azampour

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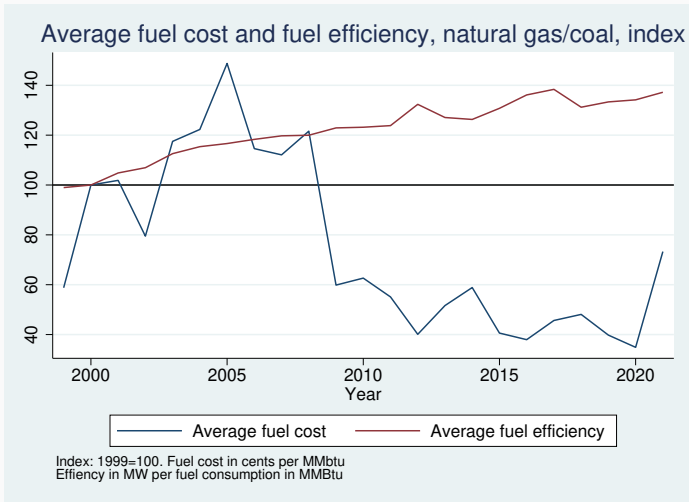
Power sector transition dynamics

- Firms' technology adoption decisions determine the pace of the green transition
- Their initial technological capital likely plays a role in this
- Need to analyze the impacts of climate policy through the lens of firm investment choices
- Climate policies impact firms' investment choices via
 - Operation costs: carbon taxes
 - Fixed costs: installation subsidies
 - Demand side: feed-in tariffs and renewable energy mandates
- How do these impact the pace of transition?

US power sector: two transitions



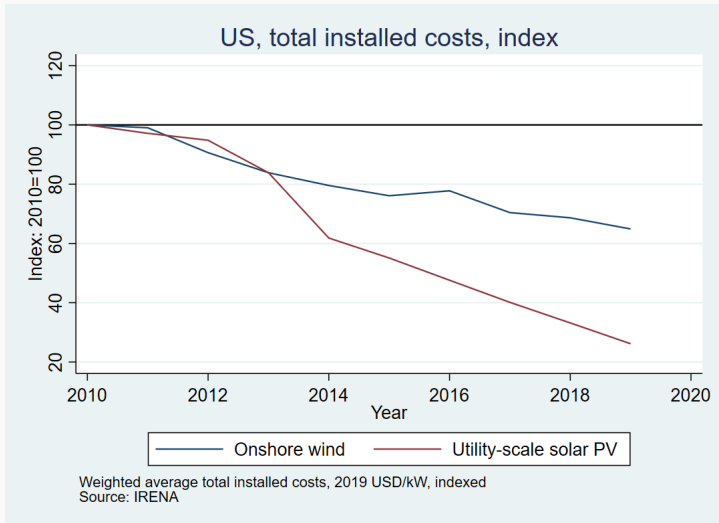
Transition 1: Conventional fuels to natural gas



Shocks in model

Price in model

Transition 2: Conventional fuels to solar and wind



”What determines the pace of technological transition in the power sector?”

- Primary goal: analyze the effects of power sector climate policies
- In addition, a micro-founded alternative for canon CES models

Outline of the project

- We present a heterogeneous agent power sector model, with generator-level technology adoption decisions.
- Data on US power generators is used to test its implications
- The model is calibrated to match the transitions towards
 1. Natural gas
 2. Solar and wind
- Counterfactual analysis to test the implications of power sector policies.

Previous literature

- Green transition and energy policy:
Arkolakis and Walsh [2023], Gentile [2024], Shapiro and Walker [2018], Acemoglu et al. [2012], Hassler et al. [2012], Acemoglu et al. [2023]
- IO literature on the energy sector:
Acemoglu et al. [2017], Cicala [2022], MacKay and Mercadal [2021]
- Technology adoption and productivity:
Perla and Tonetti [2014], Benhabib et al. [2021]
- Firm turnover and structural change:
Hopenhayn [1992], Luttmer [2007]

Patterns in the data: technology adoption

1. A generator's efficiency relative to the market declines over time, in the absence of technological updates.
2. A generator's efficiency is persistent across periods and technology updates.
3. The more time that has passed since the previous update, the higher the likelihood of an update.
4. The higher the efficiency after the previous update, the lower the likelihood of an update.
5. Generators with higher persistent quality depreciate slower.

The model

- Generators produce in a competitive market using a fuel input.
- They belong to one of two technology types: old or new.
- The productivity of each generator depends on the effective age and efficiency of their technology.
- In each period, a generator can choose to pay an adoption cost and update to a new efficiency value. The value is stochastic but known before paying the cost.
- A generator exits if its present value goes to zero.
- Entrants can pay an entry cost and draw an initial efficiency value $a_{i,0}$ in either technology.
- Entry and exit is the main margin for transition from the old to new technology.

The model: market structure

- Generators use the fuel input e_i^o or e_i^n at prices p^o or p^n , respectively
- Both types sell electricity at the price p_E
- Productivities are drawn from a normal distribution with mean $\bar{a}^o$ or $\bar{a}^n$ with the same variance σ : productivity growth is exogenous
- The two types pay technology adoption costs c_a^o and c_a^n and operation costs c_f^o and c_f^n
- Electricity demand is given by: $\ln(d_E) = \ln d_0 - \varepsilon_E \ln(p_E)$
- The supply of fuel input j is given by: $\ln e_s^j = \ln e_s^{0,j} + \epsilon_s \ln p^j$

The model: generator's period problem

- The generator's period profits are given by:

$$p_E \frac{a_i^{1+\gamma t}}{(1+g_a)^t} e_i^\alpha - p_e e_i - c_f$$

- Where $\alpha < 1$ and $\gamma > 0$
- The term:

$$\frac{a_i^{1+\gamma t}}{(1+g_a)^t}$$

is the generator's efficiency relative to the market

- g_a is the rate at which a generator's relative efficiency declines

Intra-period problem

Alternative model

Calibration

Coal-gas switching

The model: generator value function

- The value function is given by:

$$v(a_i, t) = \max_{e_i, ad=\{0,1\}} pE \frac{a_i^{1+\gamma t}}{(1+g_a)^t} e_i^\alpha - p_e e_i - c_f \\ + \beta (\mathbf{1}_{\{ad=0\}} \mathbb{E}[v(a_i, t+1)] + \mathbf{1}_{\{ad=1\}} [\mathbb{E}[v(a'_i, t-\tau)] - c_a])$$

- Technology draws follow $a'_i \sim \rho a_i + (1-\rho)\epsilon_i$ where ϵ_i follows an *i.i.d* normal distribution with the same parameters as of the entrants' distribution
- τ is the reduction in technological age following an update

The model: entry and exit

- The value of entry is given by:

$$E[v(a_{i,0}, 0)] - c_e = \int \{p_E a_{i,0} e_i^\alpha - p_e e_i + E[v(a_{i,0}, 1)]\} dD(a_{i,0}) - c_e$$

where c_e is the entry cost and $a_{i,0}$ is drawn from the exponential distribution

- If $v < 0$ the generator exits

The Equilibrium consists of policy function $P(a_i, t)$, a stationary distribution of generators Γ with the measure m_{ss} , and an equilibrium price p_E where:

- $P(a_i, t)$ is the answer to firms' problem at the price p_E
- The electricity market clears at the price p_E
- The fuel input markets clear at prices p_e^o and p_e^n
- The value of entry is zero, or the the mass of generators is zero for both technology types

Alternative model

Data: EIA power generator data

- EIA US power generator data from years 1991-2021.
- Utility power plants until year 2000: all power plants included thereafter.
- Unbalanced panel.
- Generator-level data on capacity, energy sources, operating status, planned modifications, first year of operation, location, owner, and power plant.
- Plant-level data on net generation and fuel consumption.
- The data analysis only covers operating generators that use a fuel.

Variables constructed for analysis

- Efficiency: fuel consumption per unit of net generation.
- Efficiency update dummy: 1 when efficiency increases by 5% or more but the ratio between capacity and generation does not change by more than 10% (we test several thresholds).
- Technological age variable: number of years since the latest update.
- Plant's persistent quality: average technological age at the time of update.

Efficiency

Technological age

Data: Technology adoption patterns

Table 1: Update dummy regressed on covariates, single-generator plants

VARIABLES	(1) Raw data	(2) Fixed effects
Relative efficiency after latest upgrade	-0.0630*** (0.0123)	-0.248*** (0.0916)
Years since latest update	-0.00432*** (0.000760)	0.00738*** (0.00164)
Constant	0.156*** (0.0149)	0.308*** (0.0933)
Observations	3,356	3,242
Adjusted R-squared	0.014	0.054

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Note: Update dummy defined as efficiency increase of at least 5% with less than 10% change in the capacity-to-generation ratio

Calibration, first steady state in coal to gas transition

Externally calibrated			
Param.	Description	Value	Source
$c_{o,e}/c_{n,e}$	Construction cost	2.8	Overhead cost, EIA ¹
$c_{o,f}/c_{n,f}$	Fixed operating cost	2	Fixed O&M cost, EIA ¹
$f(\epsilon_{o,i})$	Draw distribution, coal	$\mathcal{N}(0.3, 0.12)$	New plant efficiency values ²
$f(\epsilon_{n,i})$	Draw distribution, gas	$\mathcal{N}(0.4, 0.12)$	New plant efficiency values ²
ρ	Draw autocorrelation	0.7	Plant efficiency values
γ	Quality persistence	0.0075	Depreciation estimates ²
g_t	Obsolescence rate	0.015	Depreciation estimates ²
ϵ_i	Coal/gas supply elasticity	0.4	Arora and Vipin, 2014 ³
ε_E	Electricity demand elasticity	1	Burke and Abayasekara, 2018 ³
β	Depreciation rate	0.97	
Internally calibrated			
Param.	Description	Value	Matching Moment
α	Prod. function curvature	0.7125	Input use ratio $\bar{e}_o/\bar{e}_n$
$\frac{c_{o,a}}{c_{n,a}}, \frac{c_{o,e}}{c_{n,e}}$	Adoption cost ratio	1, 9.6%	Mean effic. and capacity ratio
$e_{s,0}^o, e_{s,0}^n$	Baseline coal/gas supply	2.166, 0.246	Input prices
d_0	Baseline electricity demand	9.24	Electricity price
Exit rate	Plant exit rate	1%	Plant retirement rate
τ	Technological age reduction		mean of years since update

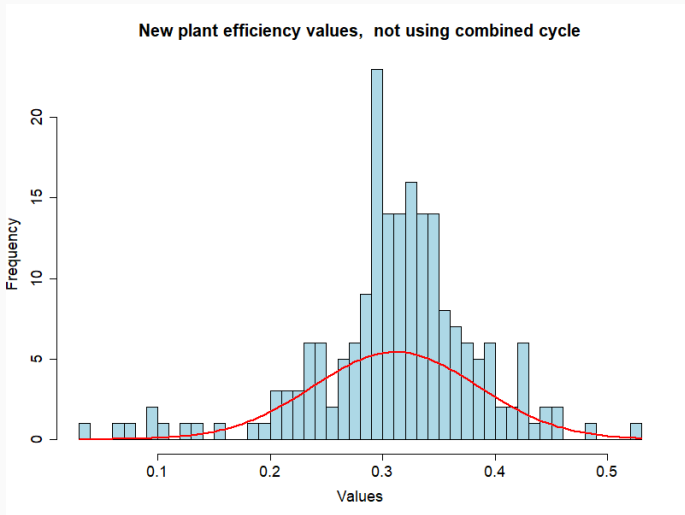
¹ USD/kW, 1997-1999 average from annual "Assumptions to the Annual Energy Outlook"

² We back out γ from a regression of efficiency values in periods between updates on covariates:

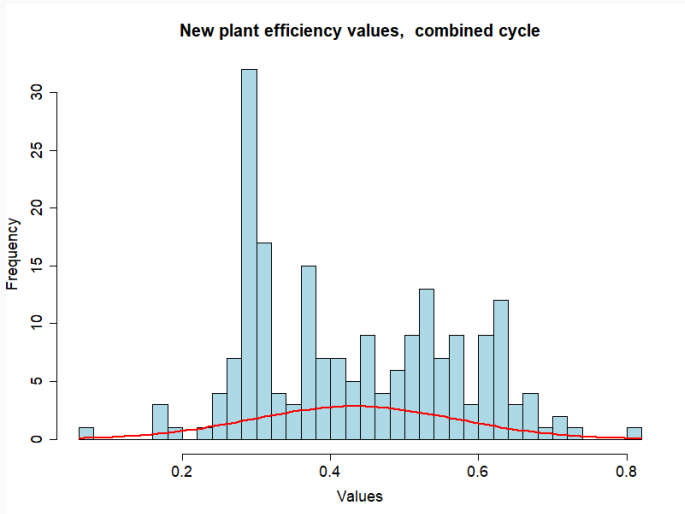
Estimation

³ Long-run elasticity estimates

Calibration: efficiency distribution, old technology

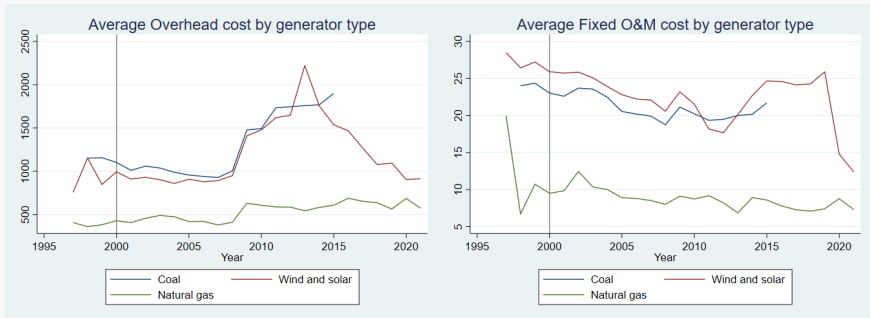


Calibration: efficiency distribution, new technology



Calibration: Construction and fixed operating costs

Figure 1: Overhead and fixed operating costs



Moments, first steady state in coal to gas transition

Moments, first steady state				
Param.	Description	Source	Data	Model
cap_o/cap_n	Measure ratio	Installed capital ratio, coal/gas	3.03	5.4
$\bar{a}$	Mean efficiency	Plant efficiency values, coal/gas	0.33	0.39
$\bar{e}_o/\bar{e}_n$	Input use ratio	Coal and gas plant fuel cons.	7.27	8.27
p^o	Input price, old	Coal price ¹	1.45	1.82
p^n	Input price, new	Natural gas price ¹	2.32	2.14
p_E	Electricity price	Retail price, industrial clients ¹	14.06	14.97

¹USD/MMBtu

Steady state results

Figure 2: Coal power plants: (relative) efficiency distribution before transition

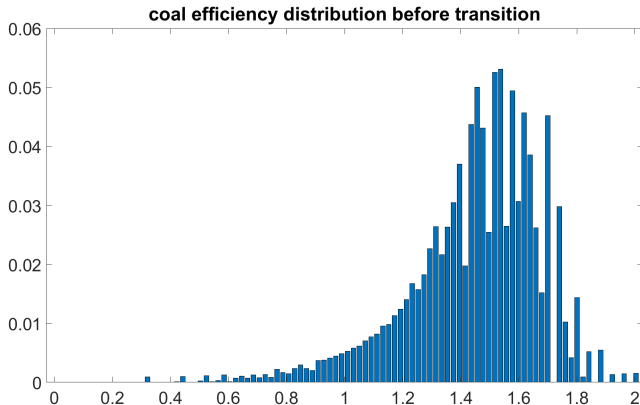
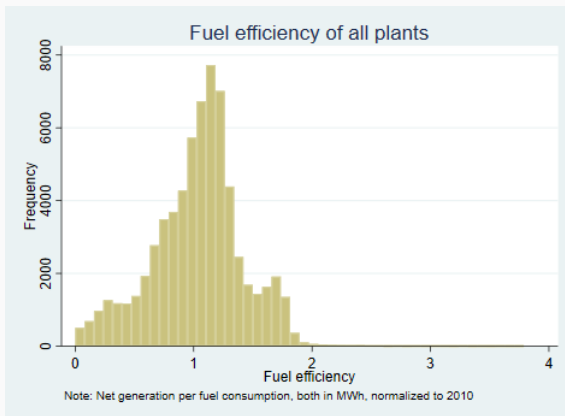


Figure 3: Relative efficiency values of all plants, normalized

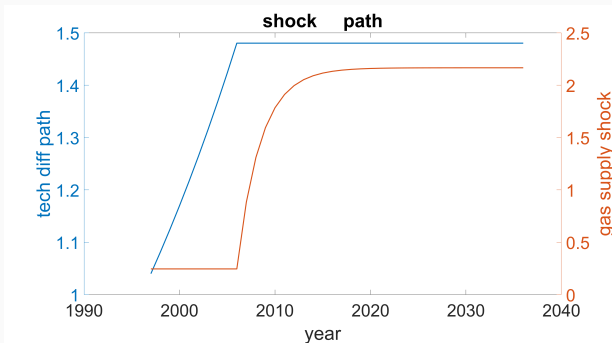


Note: Technological update implies an efficiency increase of at least 10 %, with less than 5 % variation in generator capacity-to-generation ratio

Transition 1: Coal to natural gas

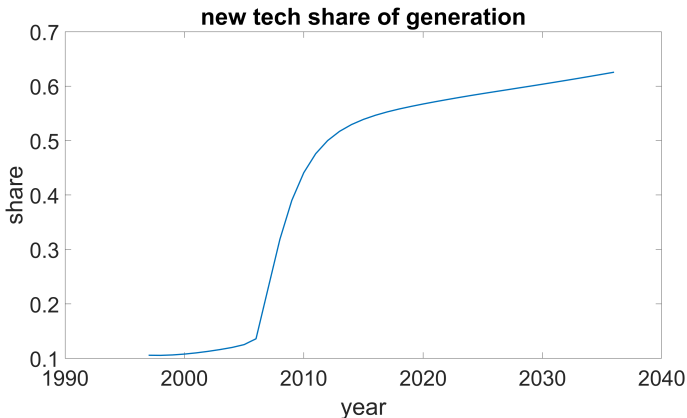
- Baseline electricity demand is growing at 1.3% per year
- Adjustment cost applied to electricity and input price changes
- Coal-to-gas transition cost equal to 40% of natural gas entry cost (EIA)

Figure 4: The path of shocks in the model



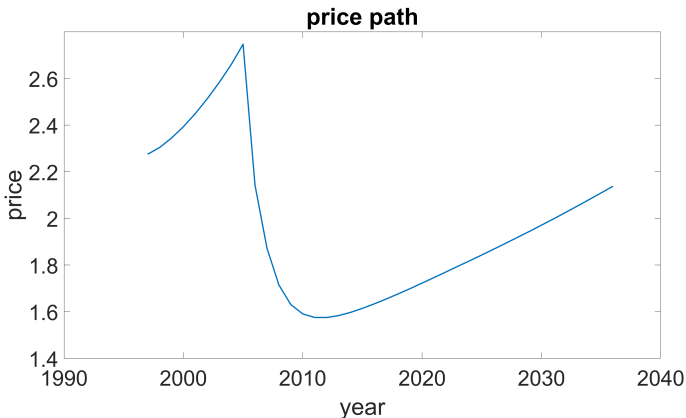
Transition 1: Coal to natural gas

Figure 5: Share of gas generation over the convergence path



Transition 1: Coal to natural gas

Figure 6: Gas price over the convergence path



Natural gas/coal cost

Next steps: policies

- Federal policies
 - Federal support to renewable energy was estimated at \$15.6 billion in 2022, up from \$1,417 million in 1999: mostly from a production tax credit
 - Support for natural gas and coal is much lower and has not increased.
- State-level policies
 - CO2 emission trading in California and ten Northeastern states
 - Renewable portfolio standard (RPS)
 - Feed-In Tariff Programs

Next steps

- Find a way to calibrate the impacts of US power sector policies
- Calibrate and solve the second transition towards solar and wind
- Counterfactual analysis
 - The impact of subsidies to solar and wind
 - The level of a carbon tax with comparative impact

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Appendix

Definitions

- The unit of our model is a power generator: an electricity production unit such as a natural gas steam turbine or a wind turbine.
- The size of a generator is not uniform.
- A power plant is a cluster of one or more generators and auxiliary equipment.
- A power sector firm may own several power plants.
- We assume perfect competition so generator decisions are independent of which power plant and firm they belong to.

Intra-period problem

From solving the generator intra-period problem we get:

$$e_i^* = \left(\frac{a_{i,t} p_E^\alpha}{p_e} \right)^{\frac{1}{1-\alpha}}$$

where

$$a_{i,t} = \frac{a_i^{1+\gamma t}}{(1+g_a)^t}$$

- In the data $a_{i,t}$ is the relative efficiency of the generator
- From the generator's perspective it faces the same problem if its own efficiency depreciates or if other generators' efficiency grows with rate $1+g$
- The latter implies that market base demand grows with rate $(1+g)^{\frac{1}{1-\alpha}}$
- We therefore incorporate depreciation and growth in the average productivity of the market in the same depreciation term

Alternative model setup

- We could also set up a generator problem with a CRS function where the average price that a generator observes is declining in production
- This is due to daily variation in electricity prices and the imperfect substitutability of generators
- As a generator increases its production this will occur at times with lower electricity prices p_E . The average price that the generator observes is:

$$p_E(e_i) = p_0 e_i^{-\eta}$$

The period problem becomes:

$$\begin{aligned} \max_{e_i} a_{i,t} e_i p_E(e_i) - p_e e_i &\rightarrow e_i^* = \left(\frac{p_0 a_{i,t} (1 - \eta)}{p_e} \right)^{\frac{1}{\eta}} \\ \implies \alpha = 1 - \eta, p_E = p_0 \frac{\eta}{1 - \eta} &\text{ equalizes the two settings} \end{aligned}$$

The model: switch from coal to gas

- Some fossil fuel generators can be converted from using one fuel to another
- In the coal-gas transition we allow coal generators to switch to natural gas by paying a switching cost
- The generator keeps it's initial productivity adjusted by the productivity gap between coal and natural gas

$$v_{c,conv}(a_i, t) = \max \left\{ v_c(a_i, t), v_g\left(\frac{a_i}{(1 + \Delta_g)^{\Delta_t}}, t\right) - c_{conv} \right\}$$

Where Δ_g is the difference between the growth rate of gas and coal efficiency idiosyncratic distributions and Δ_t is the number of years this differential growth rate has been in place.

Period problem

Value function

Why relative efficiency matters

- Let's assume that the generators depreciate with rate g_0 and the distribution of the newcomers' efficiency is growing with the rate g_n
- c_f , c_a should also grow with the rate $g_n^{\frac{1}{1-\alpha}}$: otherwise they become negligible compared to operating costs, as profits are given by:

$$\pi_{ent,t+\hat{t}}(a_j) = \left(\frac{p_E a_j (1 + g_n)^{\hat{t}}}{p_e^\alpha} \right)^{\frac{1}{1-\alpha}} \frac{1 - \alpha}{\alpha} - c_{f,t}$$
$$\pi_{update}(a_i, t) = \left(\frac{p_E a_i^{1+\gamma(t-\tau)}}{p_e^\alpha (1 + g_0)^{(t-\tau)} (1 + g_n)^\tau} \right)^{\frac{1}{1-\alpha}} \frac{1 - \alpha}{\alpha} - c_{f,t},$$

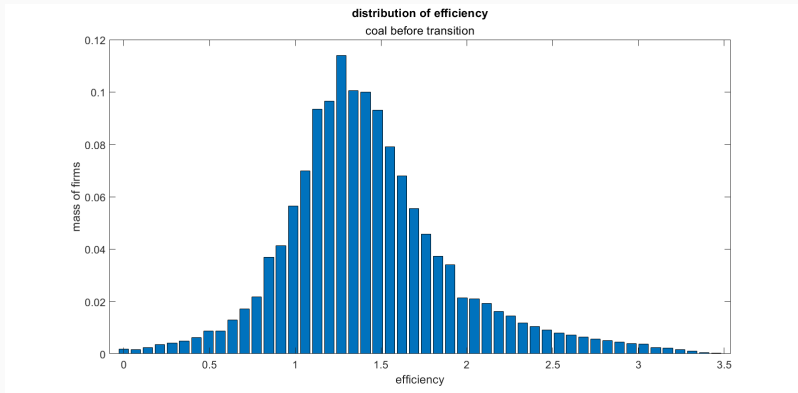
$$(\tau > t \rightarrow \tau = t)$$

Why relative efficiency matters

- This means unless $c_{f,t}$ is growing with the rate $g_n^{\frac{1}{1-\alpha}}$, in the long term it has a negligible effect on the firm's profit.
- The profits of the updating and entrant firms grow at rate g_n . This implies that $c_{a,t}$ must grow at the rate $g_n^{\frac{1}{1-\alpha}}$ for there to be a stable solution to the firm's adoption problem
- $c_{e,t}$ also needs to grow at the rate g_n so the mass of firms is constant in steady state.
- In that case we can normalize all values by the growth rate $(1 + g_n)^{\frac{1}{1-\alpha}}$.
- This means that the value of entering and adopting firm doesn't grow but the effective efficiency of non-adopting firms depreciates at the rate $1 + g_a = (1 + g_0)(1 + g_n)$
- Alternatively we can set $g_n = 0$ from the beginning and consider the effect of growth in absolute efficiencies

Model: Steady state results

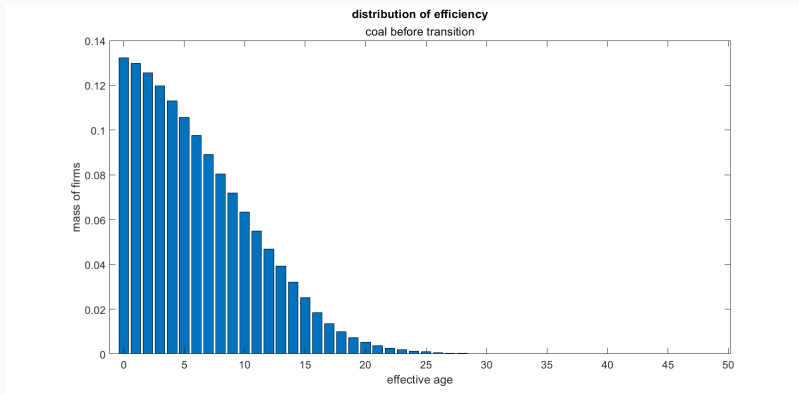
Figure 7: Coal power plants: efficiency distribution before transition



[More ...](#)

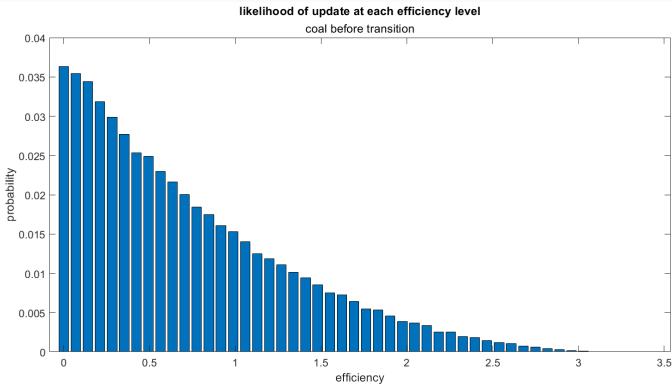
Model: Steady state results

Figure 8: Coal power plants: technological age distribution before transition



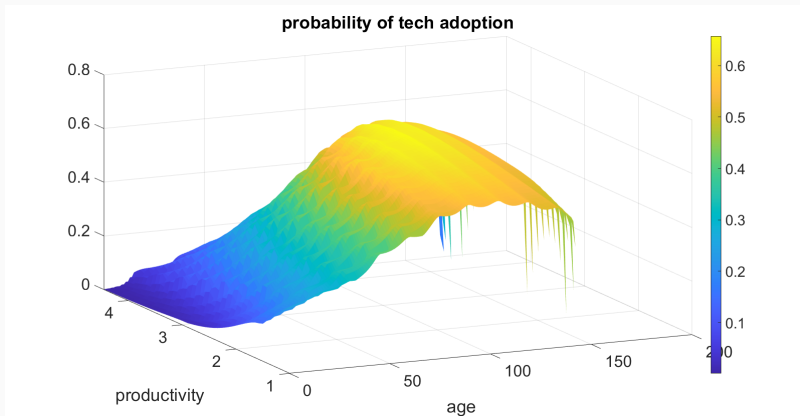
Model: Steady state results

Figure 9: Coal power plants: likelihood of technological update



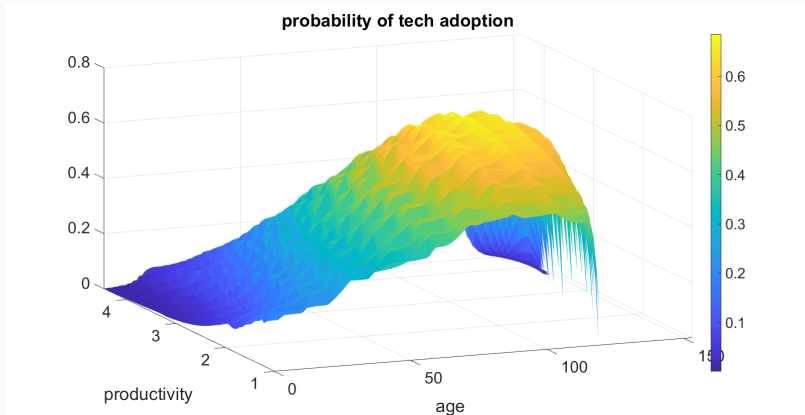
Model: Steady state results

Figure 10: Conventional power plants: probability of technology adoption before transition



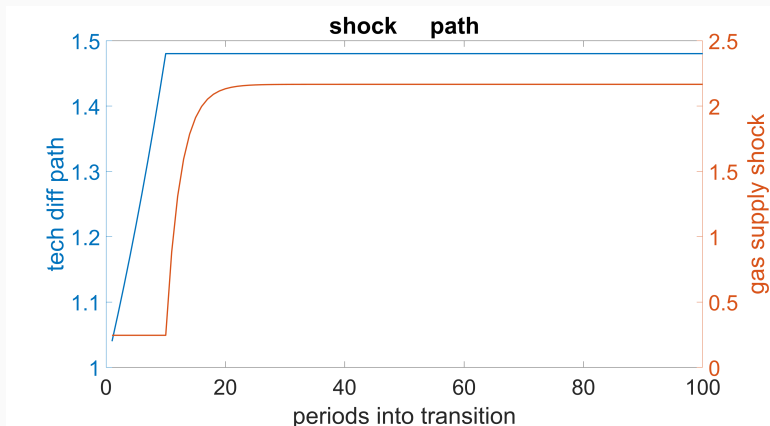
Model: Steady state results

Figure 11: Solar and wind power plants: probability of technology adoption before transition



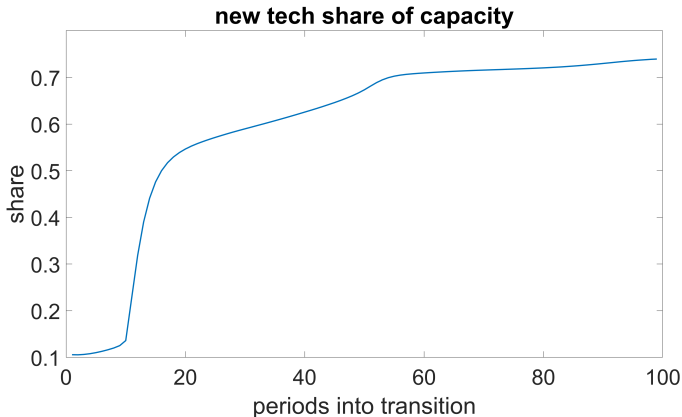
Transition 1: Conventional fuels to natural gas

Figure 12: The path of shocks in the model



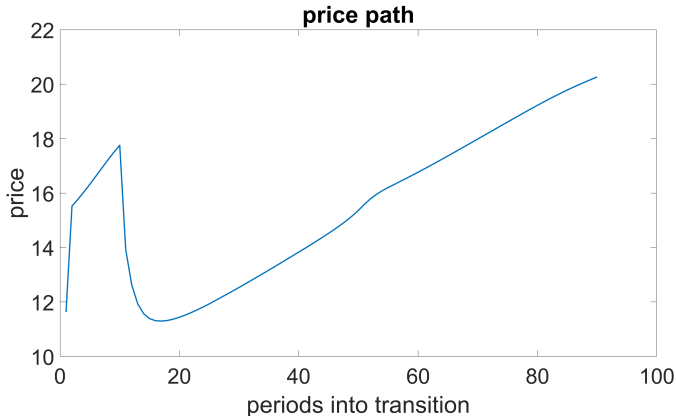
Transition 1: Conventional fuels to natural gas

Figure 13: Share of gas generation over the convergence path



Transition 1: Conventional fuels to natural gas

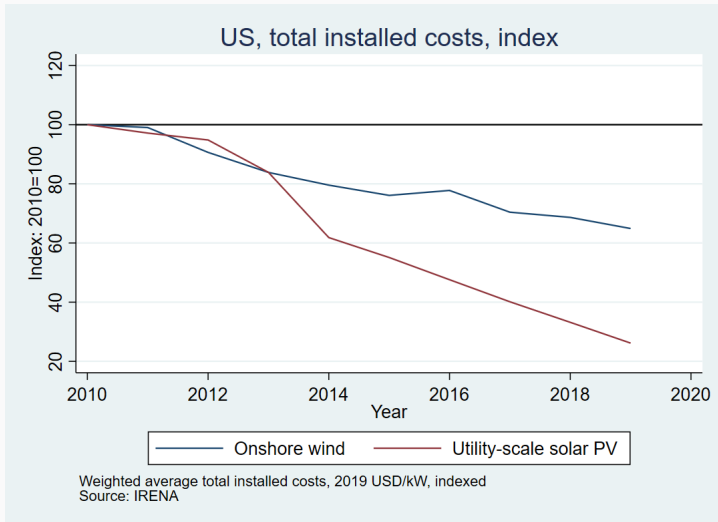
Figure 14: Electricity price over the convergence path



[transition2](#)

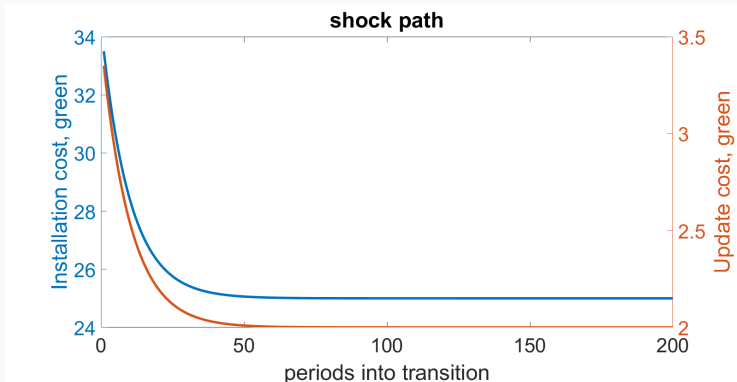
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Transition 2: Conventional fuels to solar and wind



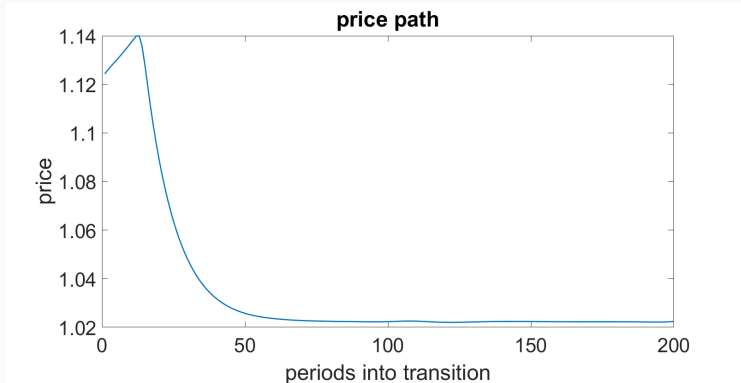
Transition 2: Conventional fuels to solar and wind

Figure 15: The path of installation costs in the model



Transition 2: Conventional fuels to solar and wind

Figure 16: Electricity price over the convergence path



transition1

Data: Adoption patterns

Table 2: Technology adoption dummy regressed on covariates, all plants

VARIABLES	(1) Raw data	(2) Fixed effects
Relative efficiency after latest upgrade	-0.0701*** (0.00653)	-0.268*** (0.0430)
Capacity weighted age	-0.00401*** (0.000356)	0.00493*** (0.000591)
Constant	0.162*** (0.00837)	0.352*** (0.0482)
Observations	13,268	12,920
Adjusted R-squared	0.017	0.061

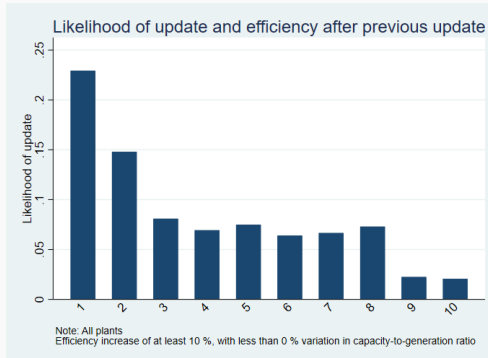
Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Note: Adoption dummy defined as efficiency increase of at least 5% with less than 10% variation in the capacity-to-generation dummy

Data: efficiency distribution

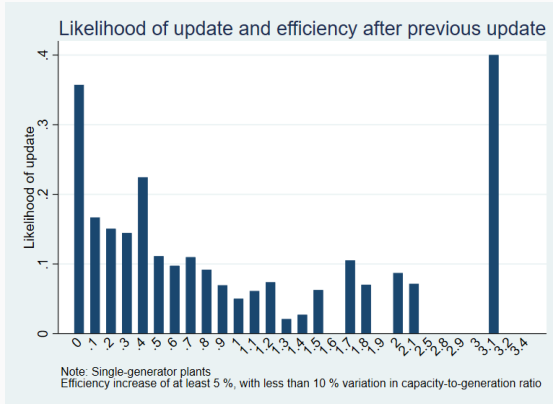
Figure 17: Number of observations, efficiency after previous update



Update pattern

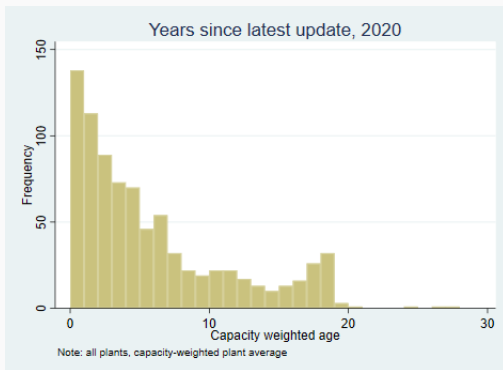
Data: Adoption patterns

Figure 18: The likelihood of update and efficiency after previous update



Years since latest update

Figure 19: Years since latest update, 2020



Note: Technological update implies an efficiency increase of at least 10 %, with less than 5 % variation in generator capacity-to-generation ratio

All efficiency values

Variables

Calibration: efficiency in periods between updates

Firm decision:

$$\begin{aligned} v(a_i, t) = & \max_{e_i, aot=\{0,1\}} p_E \frac{a_i^{1+\gamma t}}{(1+g_a)^t} e_i^\alpha - p_e e_i - c_f \\ & + \beta (\mathbf{1}_{\{aot=0\}} \mathbb{E}[v(a_i, t+1)] + \mathbf{1}_{\{aot=1\}} [\mathbb{E}[v(a'_i, t-\tau)] - c]) \end{aligned}$$

Between two updates the efficiency of a generator is then given by:

$$a_{i,t} = \frac{a_{i,t-s}}{\left(\frac{1+g_t}{a_{i,0}^\gamma}\right)^s} \rightarrow \ln a_{i,t} = \ln a_{i,t-s} - s \ln(1+g_t) + s\gamma \ln a_{i,0}$$

where s is the number of years since the latest update.

Model

Transition 1

Calibration table

Regression results: efficiency in periods between updates

Table 3: Relative efficiency in periods between updates regressed on covariates

VARIABLES	(1) Raw data	(2) Fixed effects
Years since latest update	-0.0140*** (0.00338)	-0.0316*** (0.00614)
Log relative efficiency after latest upgrade	0.954*** (0.0157)	0.459*** (0.0725)
Years since update, interaction with plant mean	0.00239*** (0.000368)	0.00277*** (0.000942)
Constant	-0.127*** (0.00981)	-0.0788*** (0.00997)
Observations	3,092	2,964
Adjusted R-squared	0.685	0.883

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Note: notes4

Efficiency in periods between updates, all plants

Table 4: Relative efficiency in periods between updates regressed on covariates

VARIABLES	(1) Raw data	(2) Fixed effects
Capacity weighted age	-0.00458*** (0.00145)	-0.0146*** (0.00159)
Log relative efficiency after latest upgrade	0.989*** (0.00807)	0.400*** (0.0626)
Years since update, interaction with plant mean	0.00110*** (0.000144)	0.000740*** (0.000173)
Constant	-0.105*** (0.00433)	-0.00677 (0.00467)
Observations	12,342	11,960
Adjusted R-squared	0.695	0.850

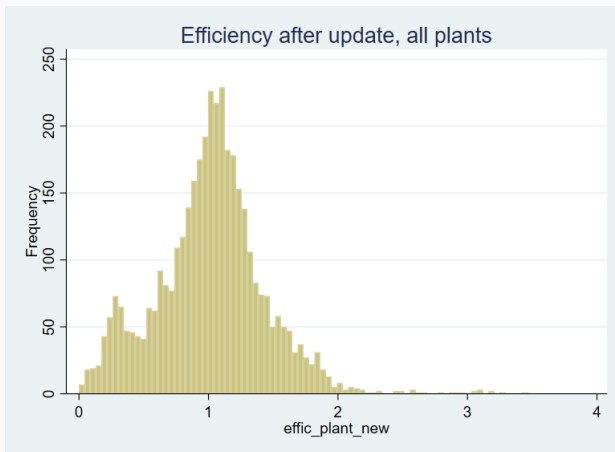
Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Note: notes4

Power plant efficiency distribution, after update

Figure 20: Distribution of efficiency values after update, all plants



Note: Technological update implies an efficiency increase of at least 5 %, with less than 10 % variation in generator capacity-to-generation ratio

Summary statistics, sample used in adoption regressions

Table 5: Summary statistics, All plants used in analysis

	mean	sd	min	max	N
No of generators	3.4	2.9	1.0	36.0	10,244
No of energy sources used	1.1	0.4	1.0	5.0	10,244
No of energy sources per generator	1.4	0.7	1.0	6.0	10,244
Average generator age	15.3	14.6	0.0	71.3	10,244
Summer capacity	262.0	414.5	0.1	3,443.0	10,244
Fuel efficiency	0.3	0.1	0.0	1.0	10,244
Relative efficiency	1.0	0.4	0.0	3.1	10,244
Relative efficiency after latest upgrade	1.1	0.4	0.0	3.3	10,244
Share updating per year	1.7	13.0	0.0	100.0	10,244
Share updating per year, unconstrained	10.0	30.0	0.0	100.0	10,244

Note: Summary statistics over plant-year

Note: Adoption dummy defined as efficiency increase of at least 10% with less than 5% variation in the capacity-to-generation dummy